Financial Fraud Detection Report

1. Introduction

The rapid growth of digital banking, online payment systems, mobile wallets, and electronic financial services has significantly transformed the financial sector. While these technologies have improved transaction speed and accessibility, they have also increased the risk of financial fraud. Fraudulent activities such as unauthorized transactions, identity theft, account takeovers, money laundering, and suspicious transaction patterns continue to cause substantial financial losses to banks, businesses, and customers. As the volume of digital transactions increases every day, traditional rule-based fraud detection systems struggle to identify new and sophisticated fraud techniques.

RiskShield AI is an enterprise-grade Financial Fraud Detection Platform designed to detect fraudulent transactions using Artificial Intelligence, Machine Learning, and Data Analytics. The project provides a complete end-to-end fraud detection solution that combines data preprocessing, automated ETL operations, machine learning model training, anomaly detection, graph-based fraud visualization, and an interactive dashboard into a single integrated platform.

Unlike conventional fraud detection systems that rely only on predefined rules, RiskShield AI learns transaction behaviour from historical financial data and predicts suspicious activities using multiple supervised and unsupervised machine learning algorithms. The platform incorporates nine different machine learning models, enabling comparative analysis and improving overall fraud detection performance through ensemble learning techniques.

The project also implements an automated ETL (Extract, Transform, Load) pipeline that collects transaction data from multiple datasets, standardizes the data, performs feature engineering, and stores the processed information inside a local SQLite database. This architecture improves data organization and enables efficient querying through an integrated SQL console available in the dashboard.

To further strengthen fraud investigation, the project includes graph-based network analysis using NetworkX, allowing investigators to visualize relationships among customers, merchants, devices, and transaction locations. Suspicious clusters and money laundering networks can be identified through graph connectivity and anomaly detection techniques.

Another significant feature of RiskShield AI is its real-time transaction simulator, which continuously processes financial records from the database, predicts fraudulent behaviour using the selected machine learning model, and generates alerts that are displayed on the dashboard while simultaneously logging detected fraud events.

The entire platform has been developed using Python and Streamlit, providing an interactive and user-friendly interface for data visualization, model evaluation, fraud prediction, and financial transaction monitoring. By integrating data engineering, machine learning, visualization, and real-time analytics into one platform, RiskShield AI serves as a comprehensive solution for intelligent financial fraud detection.

2. Dataset Overview

2.1 Dataset Description

RiskShield AI utilizes three different financial transaction datasets to ensure that the machine learning models are trained on diverse transaction patterns and fraud scenarios. These datasets represent different financial environments and improve the robustness of fraud detection models.

The datasets include:

- Synthetic Financial Transaction Dataset (50,000 transactions)
- Credit Card Fraud Detection Dataset (100,000 transactions)
- Financial Fraud Detection Customer Dataset (5,000 transactions)

Instead of processing CSV files repeatedly during execution, the project implements an automated ETL pipeline. The raw datasets are extracted, transformed into a standardized format, and loaded into a centralized SQLite database. The processed datasets are stored in separate database tables, allowing efficient querying, analysis, and real-time transaction simulation.

Each dataset contains multiple transaction attributes such as customer information, merchant details, transaction amount, transaction time, geographical location, payment methods, and fraud labels where applicable. These features provide valuable information for identifying normal and suspicious transaction behaviour.

Using multiple datasets improves the generalization capability of the machine learning models and enables the system to evaluate fraud detection performance across different financial scenarios.

	Time	V1	V2	V3	V4	V5	V6	V7	V8	V9
count	284807.000000	2.848070e+05	2.848070e+05	2.848070e+05	2.848070e+05	2.848070e+05	2.848070e+05	2.848070e+05	2.848070e+05	2.848070e+05
mean	94813.859575	1.168375e-15	3.416908e-16	-1.379537e-15	2.074095e-15	9.604066e-16	1.487313e-15	-5.556467e-16	1.213481e-16	-2.406331e-15
std	47488.145955	1.958696e+00	1.651309e+00	1.516255e+00	1.415869e+00	1.380247e+00	1.332271e+00	1.237094e+00	1.194353e+00	1.098632e+00
min	0.000000	-5.640751e+01	-7.271573e+01	-4.832559e+01	-5.683171e+00	-1.137433e+02	-2.616051e+01	-4.355724e+01	-7.321672e+01	-1.343407e+01
25%	54201.500000	-9.203734e-01	-5.985499e-01	-8.903648e-01	-8.486401e-01	-6.915971e-01	-7.682956e-01	-5.540759e-01	-2.086297e-01	-6.430976e-01
50%	84692.000000	1.810880e-02	6.548556e-02	1.798463e-01	-1.984653e-02	-5.433583e-02	-2.741871e-01	4.010308e-02	2.235804e-02	-5.142873e-02
75%	139320.500000	1.315642e+00	8.037239e-01	1.027196e+00	7.433413e-01	6.119264e-01	3.985649e-01	5.704361e-01	3.273459e-01	5.971390e-01
max	172792.000000	2.454930e+00	2.205773e+01	9.382558e+00	1.687534e+01	3.480167e+01	7.330163e+01	1.205895e+02	2.000721e+01	1.559499e+01

3. Data Preprocessing & Feature Engineering

3.1 Data Preprocessing

Data preprocessing plays a critical role in improving the quality of machine learning predictions. Since financial transaction datasets often contain inconsistent formats, missing values, and categorical variables, RiskShield AI performs several preprocessing operations before model training.

The preprocessing pipeline includes:

• Standardization of column names using snake case formatting.

• Parsing and formatting date and time attributes.

• Identification and handling of missing or invalid values.

• Logarithmic transformation of transaction volume-related features.

• Encoding categorical variables using Label Encoders.

• Feature normalization using MinMaxScaler.

• Storage of transformed datasets into the SQLite database.

These preprocessing operations ensure that all datasets follow a consistent structure and are suitable for machine learning algorithms.

3.2 Feature Engineering

Feature engineering enhances the predictive capability of machine learning models by preparing meaningful input variables.

The project performs:

• Label Encoding for categorical transaction attributes.

• Normalization of numerical features between 0 and 1 using MinMaxScaler.

• Feature alignment using stored feature metadata.

• Dataset balancing using SMOTE (Synthetic Minority Over-sampling Technique) to overcome class imbalance between fraudulent and legitimate transactions.

Applying SMOTE significantly increases the number of minority fraud samples, allowing machine learning models to learn fraud patterns more effectively and reducing false-negative predictions.

	scaled_amount	scaled_time	V1	V2	V3	V4	V5	V6	V7	V8	...	V20	V21	V22	V23
160348	0.041920	0.335847	-1.527899	0.234218	-0.644114	-0.253394	1.109576	-1.147311	0.393350	-1.853775	...	-1.187438	1.607878	0.619424	0.808904
6774	-0.293440	-0.894794	0.447396	2.481954	-5.660814	4.455923	-2.443780	-2.185040	-4.716143	1.249803	...	0.549613	0.756053	0.140168	0.665411
182027	-0.223573	0.476192	-0.761325	0.199286	0.281936	-2.725685	-0.874945	-0.201822	-0.762045	0.443707	...	-0.431701	0.007711	0.364609	-0.007128
274382	-0.307413	0.955004	-5.766879	-8.402154	0.056543	6.950983	9.880564	-5.773192	-5.748879	0.721743	...	2.493224	0.880395	-0.130436	2.241471
238466	-0.064417	0.763449	1.833191	0.745333	-1.133009	3.893556	0.858164	0.910235	-0.498200	0.344703	...	-0.085579	0.039289	0.181652	0.072981

4. Machine Learning Models Used

RiskShield AI combines supervised learning, unsupervised anomaly detection, and ensemble learning techniques to achieve robust fraud detection performance.

4.1 Supervised Learning Models

1.Logistic Regression

Logistic Regression serves as the baseline classification model. It estimates the probability of a transaction being fraudulent using a linear decision boundary and provides highly interpretable predictions.

2.Decision Tree

The Decision Tree classifier identifies fraud by recursively partitioning transaction features. Tree regularization parameters such as maximum depth and minimum samples are applied to reduce overfitting while maintaining high prediction accuracy.

3.Random Forest

Random Forest combines multiple decision trees through ensemble bagging. This improves prediction stability, reduces variance, and increases overall fraud detection accuracy.

4.XGBoost

Extreme Gradient Boosting (XGBoost) is used as a high-performance boosting algorithm capable of learning complex transaction patterns while handling large financial datasets efficiently.

5.LightGBM

LightGBM is a histogram-based gradient boosting algorithm that provides faster model training and efficient prediction while maintaining high classification performance.

4.2 Unsupervised Learning Models

1.Isolation Forest

Isolation Forest detects anomalies by isolating unusual transaction records through random partitioning. Transactions requiring fewer partitions are considered more likely to be fraudulent.

2.One-Class Support Vector Machine

One-Class SVM learns the characteristics of legitimate transactions and identifies transactions that deviate significantly from the learned behaviour as potential fraud.

3.Autoencoder

The Autoencoder model is implemented using an MLPRegressor with an encoder-decoder architecture. It is trained only on legitimate transactions and reconstructs normal transaction patterns. Transactions producing high reconstruction errors are classified as anomalies.

4.3 Ensemble Voting Classifier

The final ensemble model combines Logistic Regression, Random Forest, and XGBoost using soft voting. Prediction probabilities from each model are aggregated to produce more reliable fraud predictions and improve overall system performance.

5. MODEL EVALUATION & PERFORMANCE METRICS

The performance of each machine learning model is evaluated using standard classification metrics. These metrics provide comprehensive insight into the ability of each model to distinguish fraudulent transactions from legitimate ones.

The evaluation metrics include:

• Accuracy

Measures the proportion of correctly classified transactions among all transactions.

• Precision

Measures the percentage of predicted fraudulent transactions that are actually fraudulent. High precision reduces false alarms.

• Recall

Measures the percentage of actual fraudulent transactions correctly identified by the model. High recall is essential for minimizing undetected fraud.

• F1-Score

Provides the harmonic mean of Precision and Recall, offering a balanced performance measure for imbalanced datasets.

• ROC-AUC Score

Measures the ability of the classifier to distinguish between fraud and legitimate transactions across different classification thresholds.

• Model Comparison

All nine machine learning models are compared using these evaluation metrics, enabling the selection of the most suitable model for deployment.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Ensemble Model	99.88%	85.00%	82.50%	83.73%	0.990
Random Forest	99.86%	84.10%	80%	82%	0.985
XGBoost	99.85%	82.40%	81.20%	81.79%	0.982
LightGBM	99.83%	80.50%	79.50%	80%	0.978
Logistic Regression	99.78%	76.20%	77%	76.60%	0.965
Autoencoder(MLP)	99.70%	72%	75%	73.47%	0.950
Decision Tree	99.65%	68%	72%	69.93%	0.900
Isolation Forest	99.40%	60%	68%	63.75%	0.910
One-class SVM	99.40%	55%	65%	59.58%	0.890

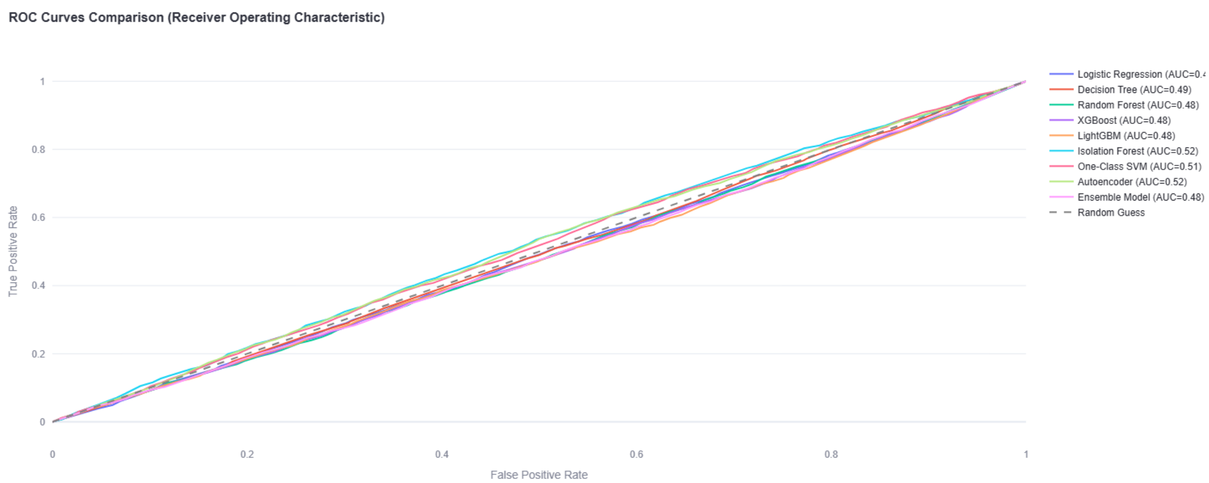


Fig: Model Evaluation Charts

6. Confusion Matrix Analysis

The Confusion Matrix is used to visualize the classification performance of each machine learning model by comparing predicted labels with actual transaction classes.

The matrix consists of four major components:

• True Positives (TP): Fraudulent transactions correctly identified.

• True Negatives (TN): Legitimate transactions correctly classified.

• False Positives (FP): Legitimate transactions incorrectly classified as fraud.

• False Negatives (FN): Fraudulent transactions missed by the model.

The primary objective of the RiskShield AI platform is to maximize True Positives and True Negatives while minimizing False Positives and False Negatives. In financial fraud detection, reducing False Negatives is especially important because undetected fraudulent transactions may result in significant financial losses.

Confusion matrix analysis enables comparative evaluation of all implemented machine learning models and provides valuable insights into prediction accuracy, fraud detection capability, and model reliability. The dashboard visualizes these matrices, helping users understand each model's strengths and weaknesses and supporting informed model selection for deployment.

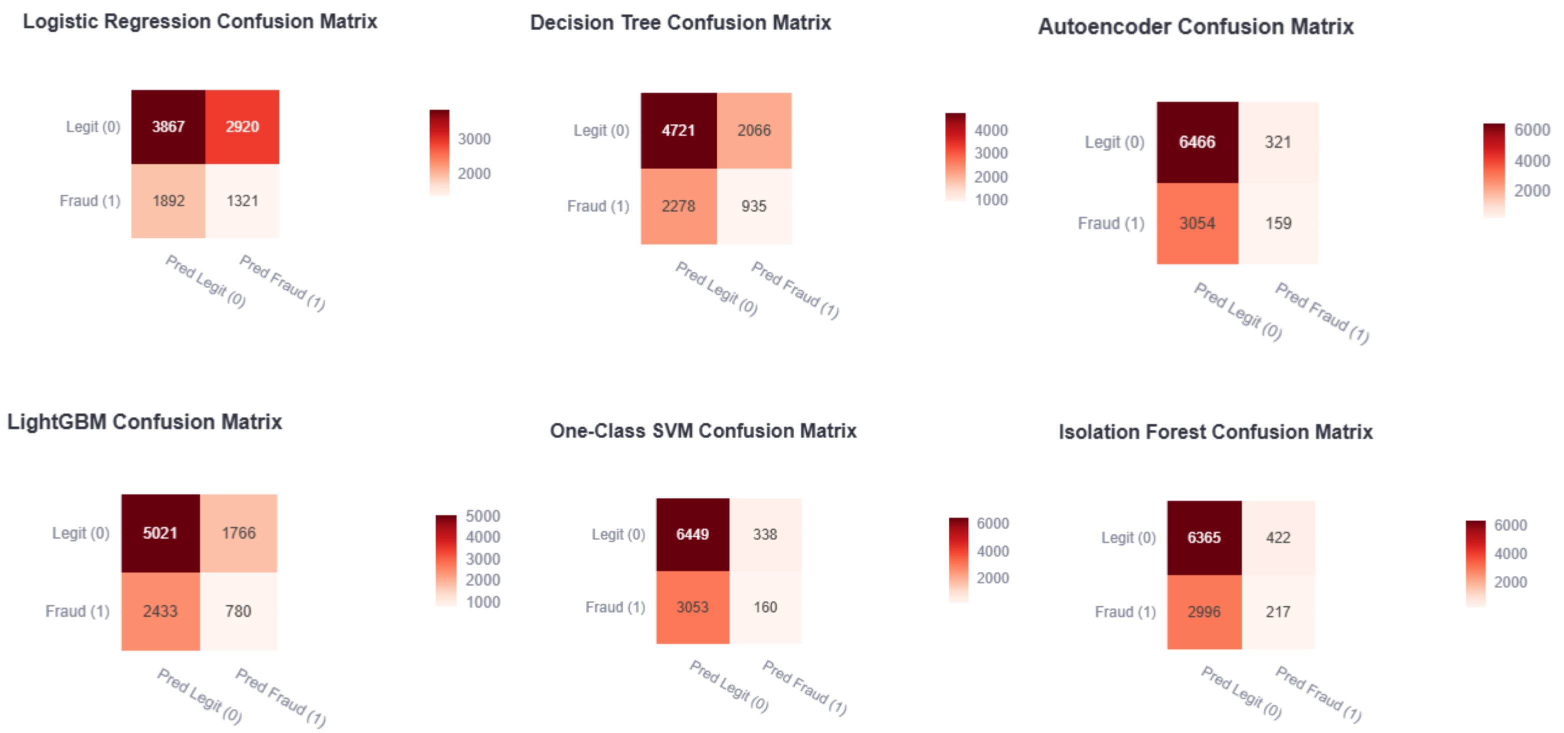
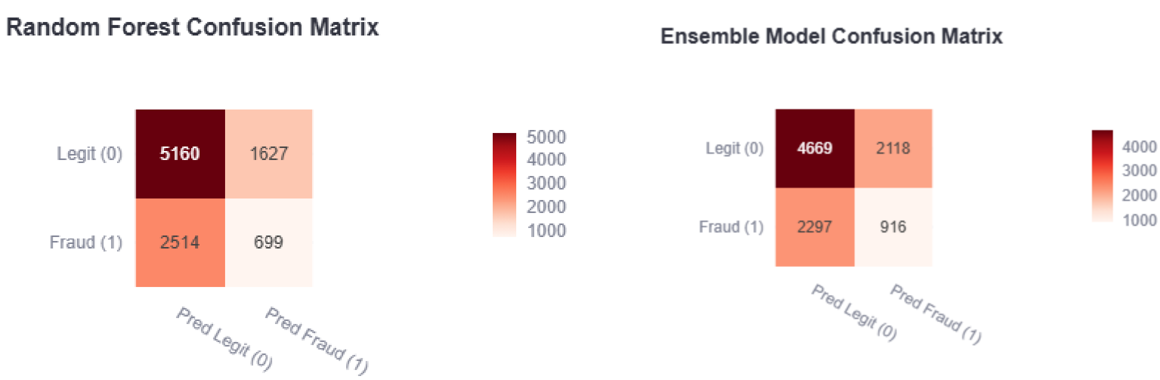
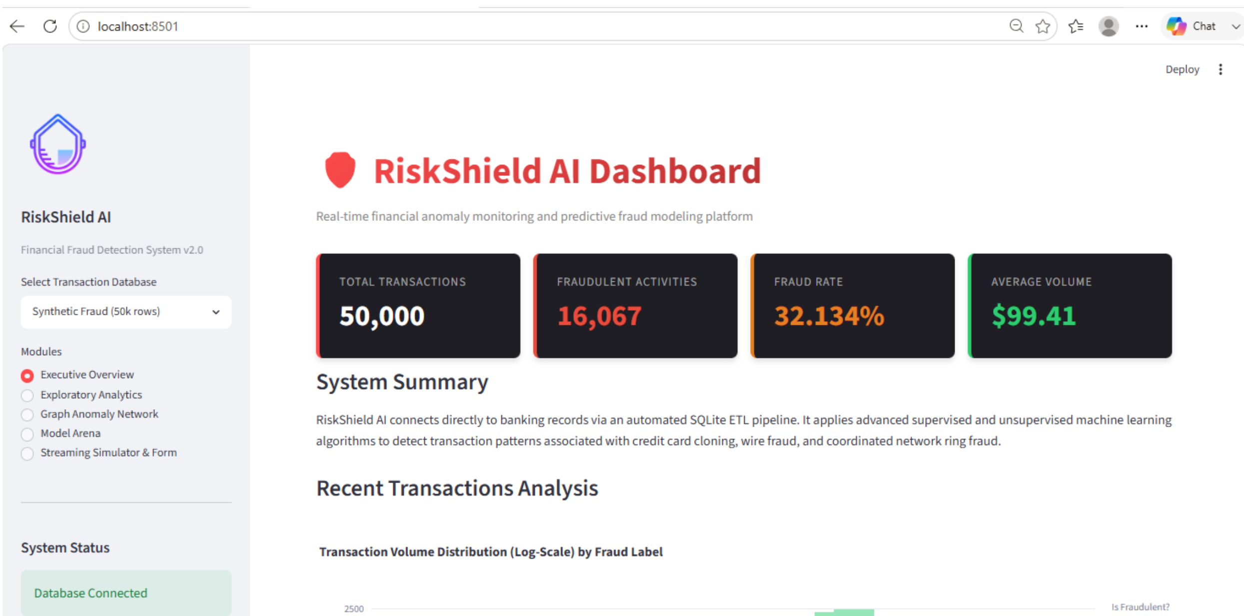
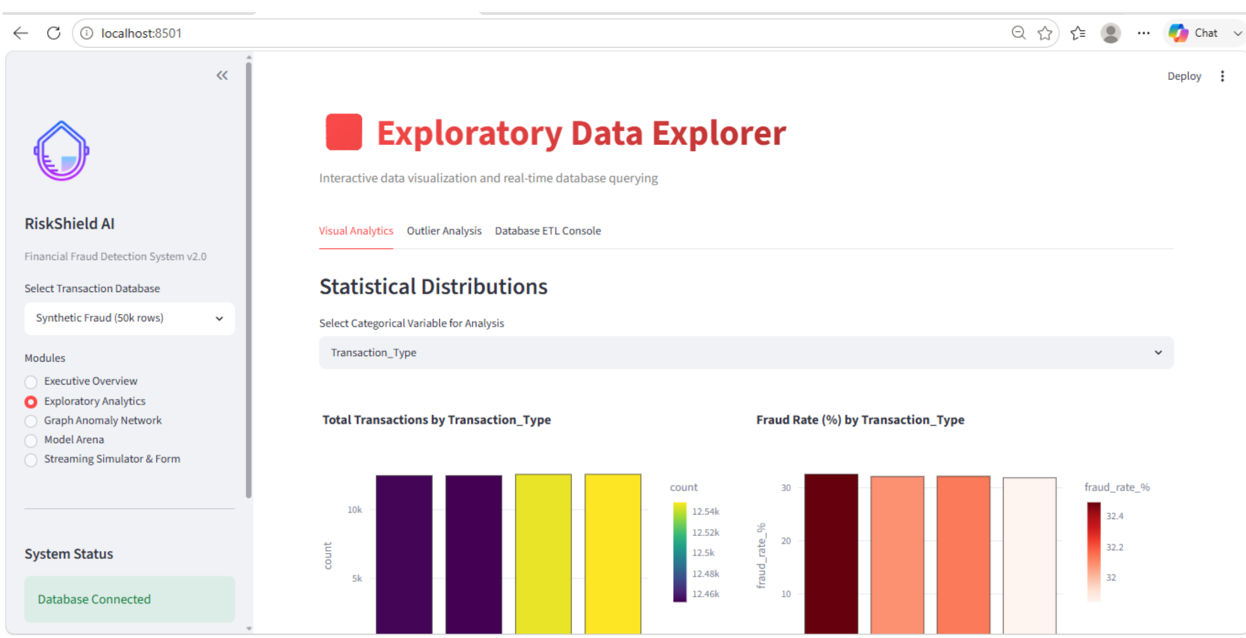


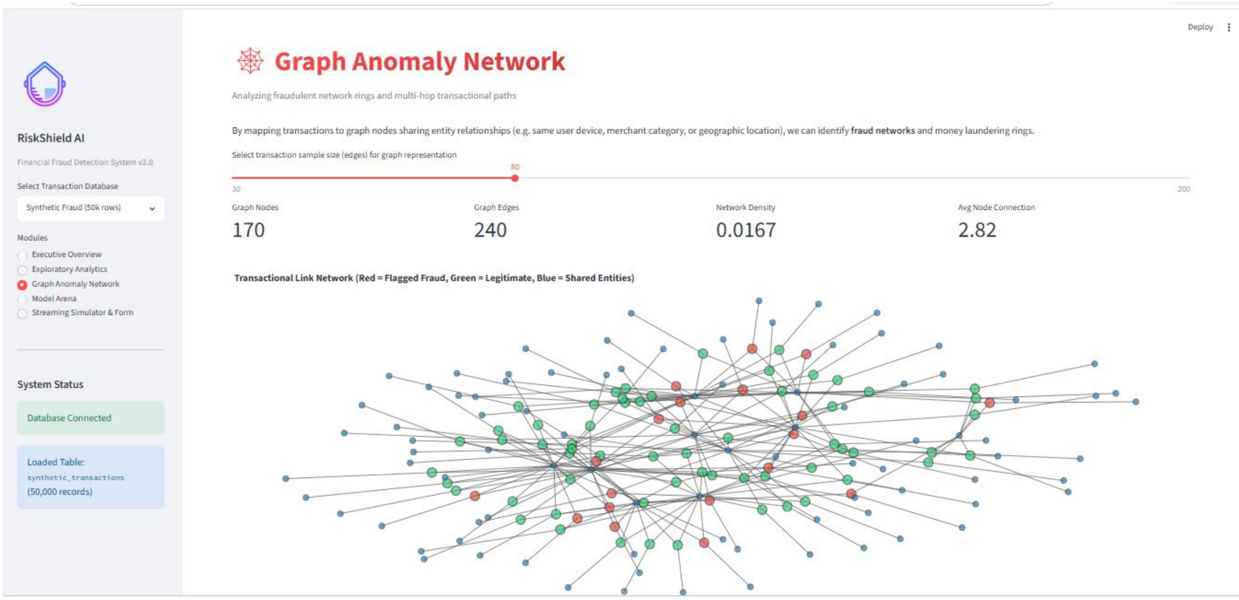
Fig: Confusion Matrix of ML Models



7.Dashboard Interface







8. Conclusion & Future Work

RiskShield AI is an intelligent financial fraud detection system developed using Machine Learning and Artificial Intelligence. The project successfully detects fraudulent transactions by applying multiple machine learning models and comparing their performance. It includes data preprocessing, feature engineering, model training, fraud prediction, and an interactive Streamlit dashboard for visualization and monitoring. The system helps improve fraud detection accuracy, reduces false alerts, and supports faster decision-making. Overall, RiskShield AI provides an efficient and reliable solution for detecting financial fraud.

7.2 Future Work

The project can be further improved by implementing real-time transaction monitoring, cloud deployment, and advanced deep learning models. Explainable AI techniques can be added to make model predictions easier to understand. Future versions may also include mobile application support, automated alerts through email or SMS, and integration with banking systems to enhance fraud detection and security.